

MRPL Metallurgical & Ultrasonic Corrosion Rate Audit — Column Vessel V-1001

Ref: MRPL/MET/2026/CR-042

Turnaround Historical Trend

Year	Measured Thickness (mm)	Trend
2018	12.16	Baseline
2022	11.68	−0.48 mm
2026	11.20	−0.48 mm

Long-term vs Short-term Corrosion Evaluation

The evaluated corrosion rate is 0.120 mm/year. Remaining-life calculations should be confirmed against approved inspection history, measurement uncertainty and applicable engineering procedures.

NDT Ultrasonic Probe Calibration Parameters

Parameter	Setting
Probe Type	Dual-element ultrasonic thickness probe
Calibration Block	Certified reference block
Couplant	Industrial UT couplant
Frequency	5 MHz
Calibration Check	Before / after inspection

Metallurgical Assessment

The thickness trend is consistent with predominantly uniform wall loss in the evaluated section. No localized naphthenic acid cracking indication was recorded in the inspection assessment.

Certified by: Rina Sharma, Quality Inspection Lead